



Abstract

Individuals who are hard of hearing often experience difficulty identifying the direction of surrounding sounds, reducing situational awareness and personal safety. To address this challenge, we developed Audiovisor, a wearable system that converts sound direction into visual feedback for the user. During prototype testing, Audiovisor correctly classified the dominant sound direction in over 84% of trials, providing real-time feedback with an effective angular resolution of approximately $\pm 45^\circ$. The system demonstrated a response latency of around 200 ms, allowing users to quickly recognize where sounds originated. These results highlight the potential of accessible, wearable audio-to-visual technologies, to improve the overall quality of life for individuals with hearing impairments.

Objectives

1. Develop a wearable audio-to-visual assistive system that indicates the direction of detected sounds using LEDs.
2. Accurately differentiate signal levels from multiple microphones to determine the dominant sound direction and activate the corresponding LED.
3. Ensure the system is portable and suitable for wearable use through compact integration and low power operation.
4. Achieve correct LED activation that is clearly visible to the user from the appropriate direction with real-time system response (latency ~ 200 ms).
5. Validate system performance through repeated testing, demonstrating close to 84% directional accuracy with reliable and consistent operation.

Materials and Methods

Materials: PDM MEMS microphones, Programmable LEDs, ESP32, glasses frame

Methods

- Capture dual-channel audio via a microphone array at a 16 kHz sampling rate, utilizing twin data buffers to segment input by orientation (Right/Back and Left/Down).
- Extract amplitude profiles of the audio by applying Root Mean Square (RMS) analysis to isolated buffer streams.
- Triggers targeted LED indicators based on dynamic thresholding to visualize the sound source's origin.
- Streams processed audio data to a web-based dashboard via Bluetooth Low Energy (BLE) for real time monitoring.
- Dashboard provides remote parameter adjustment such as LED brightness and sensitivity thresholds via BLE.

Diagrams/Figures/Experiments

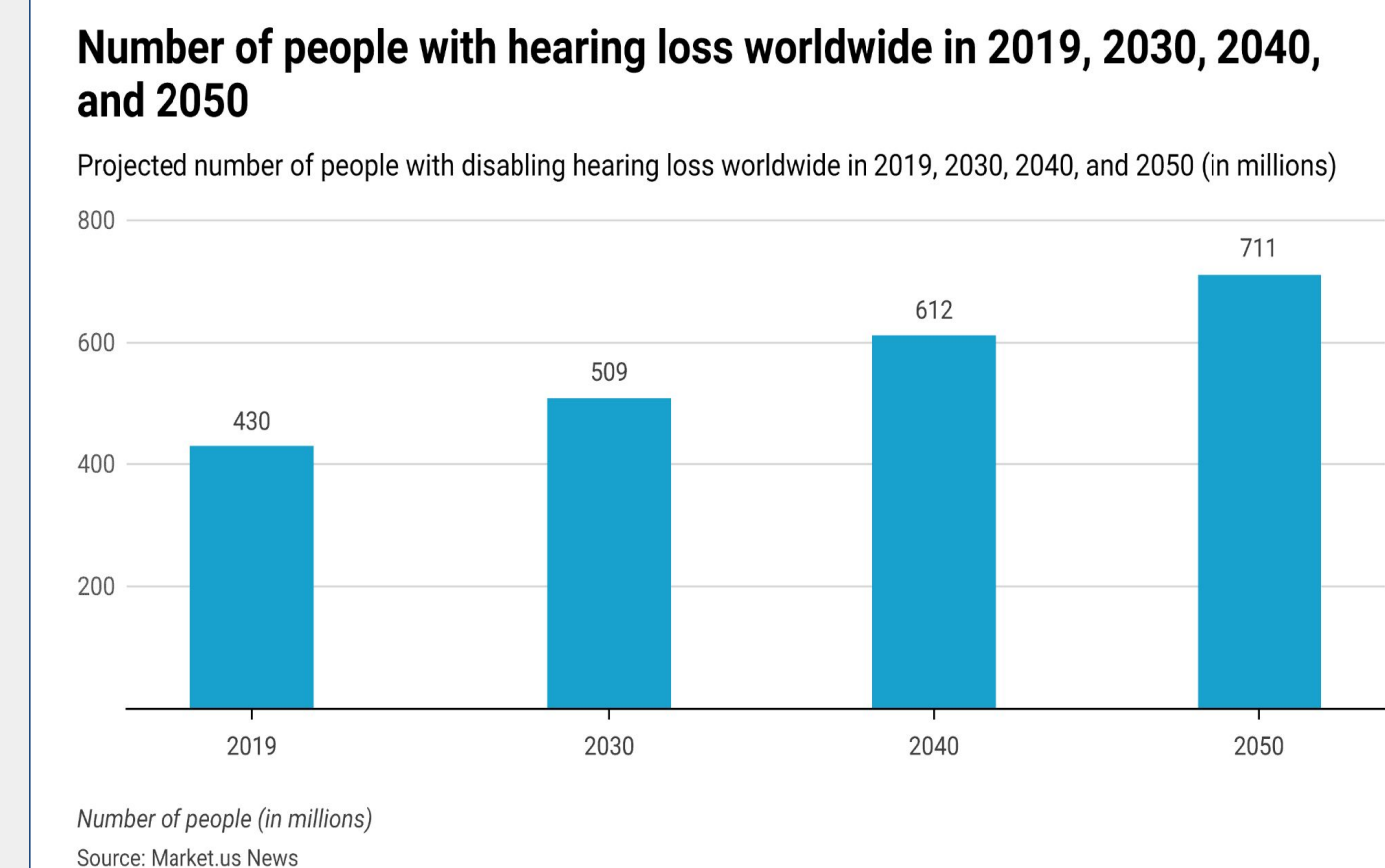
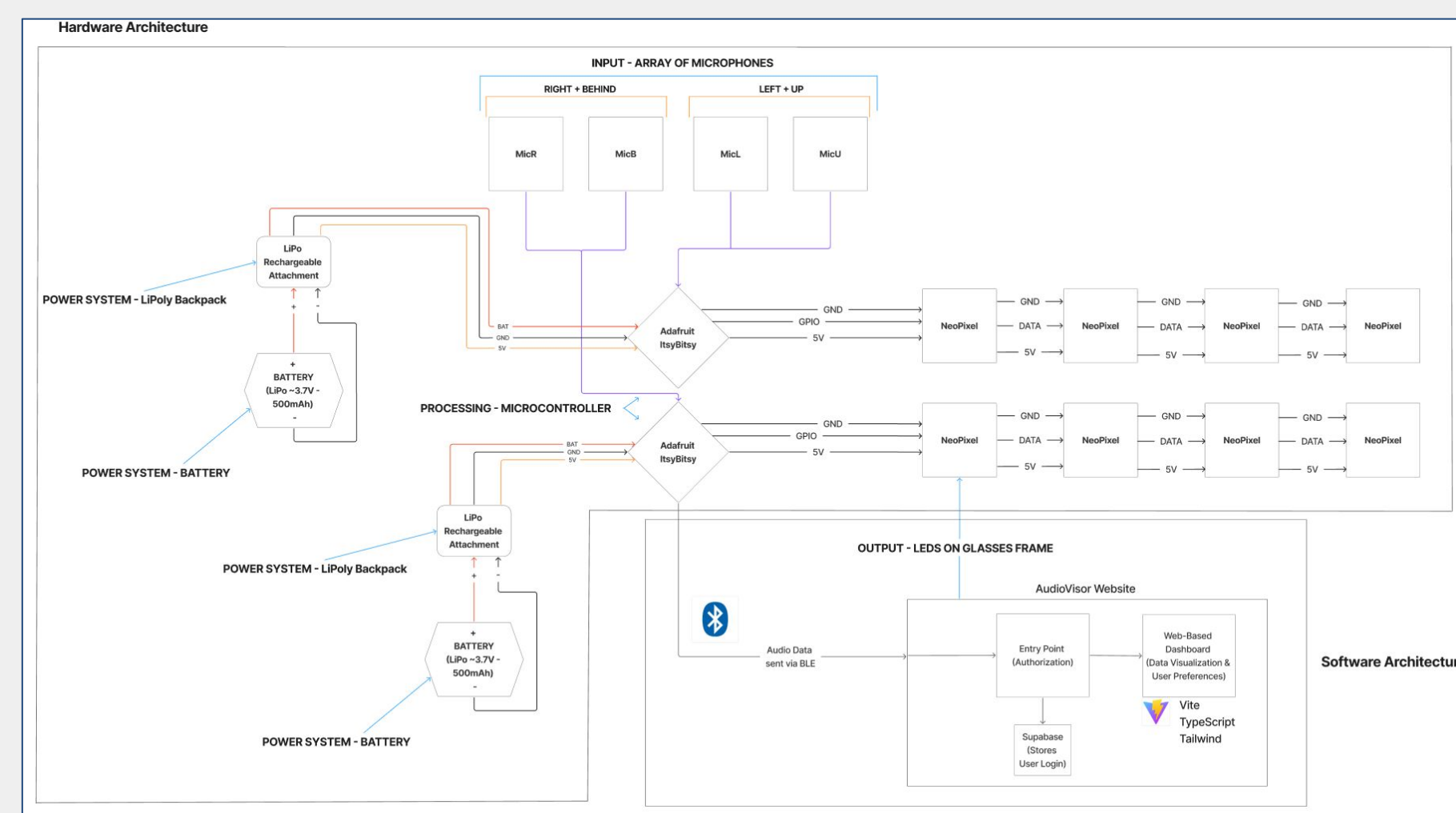
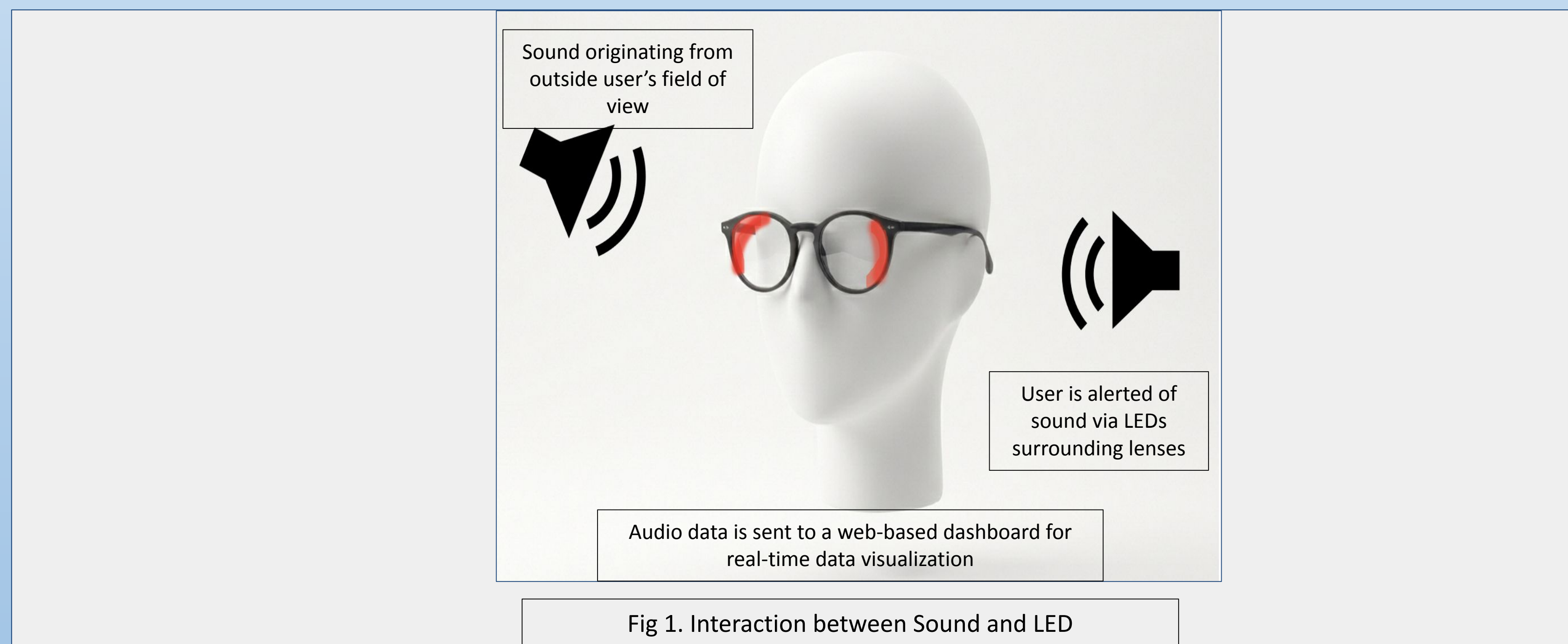


Fig 3. Global Rise in Hearing Loss

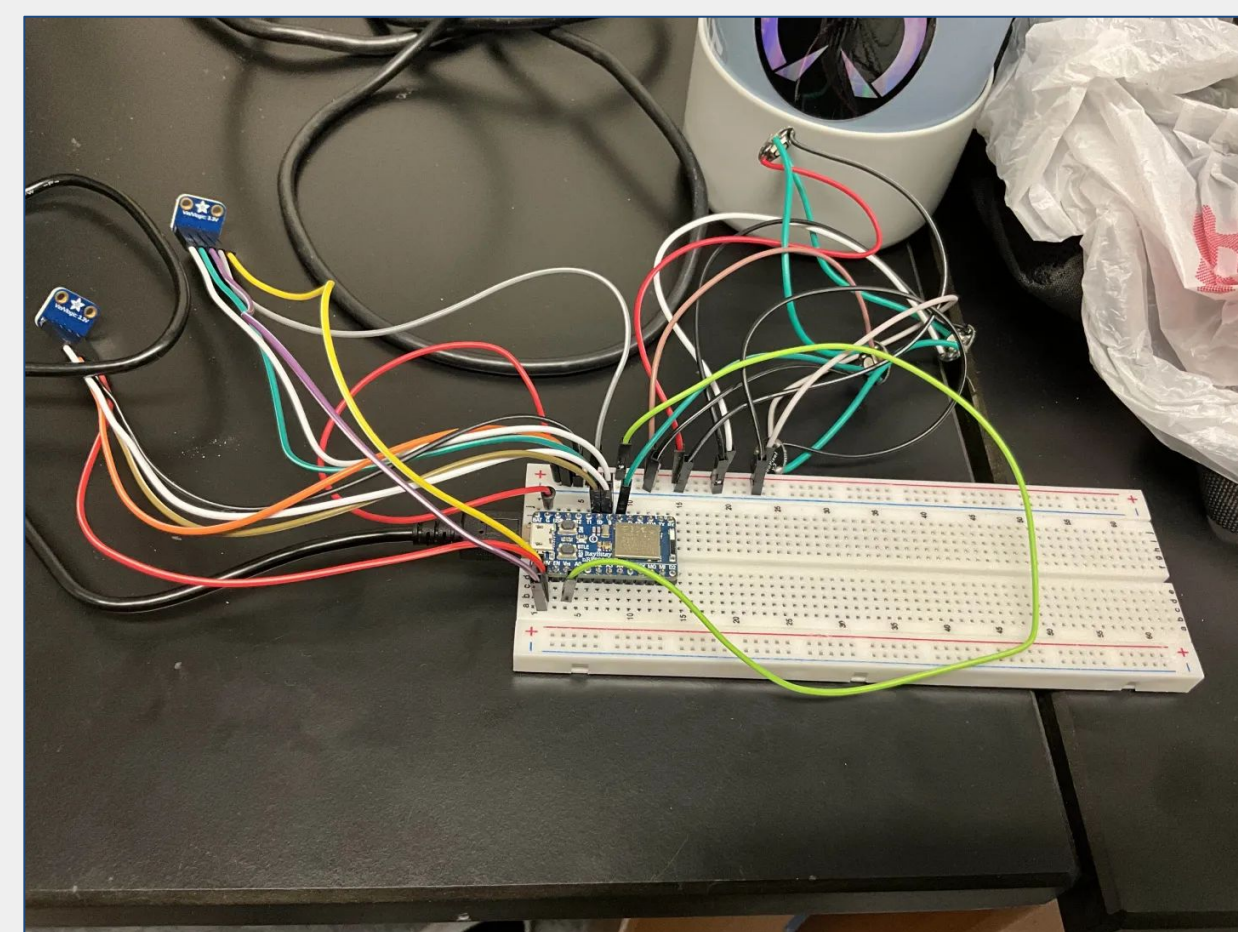
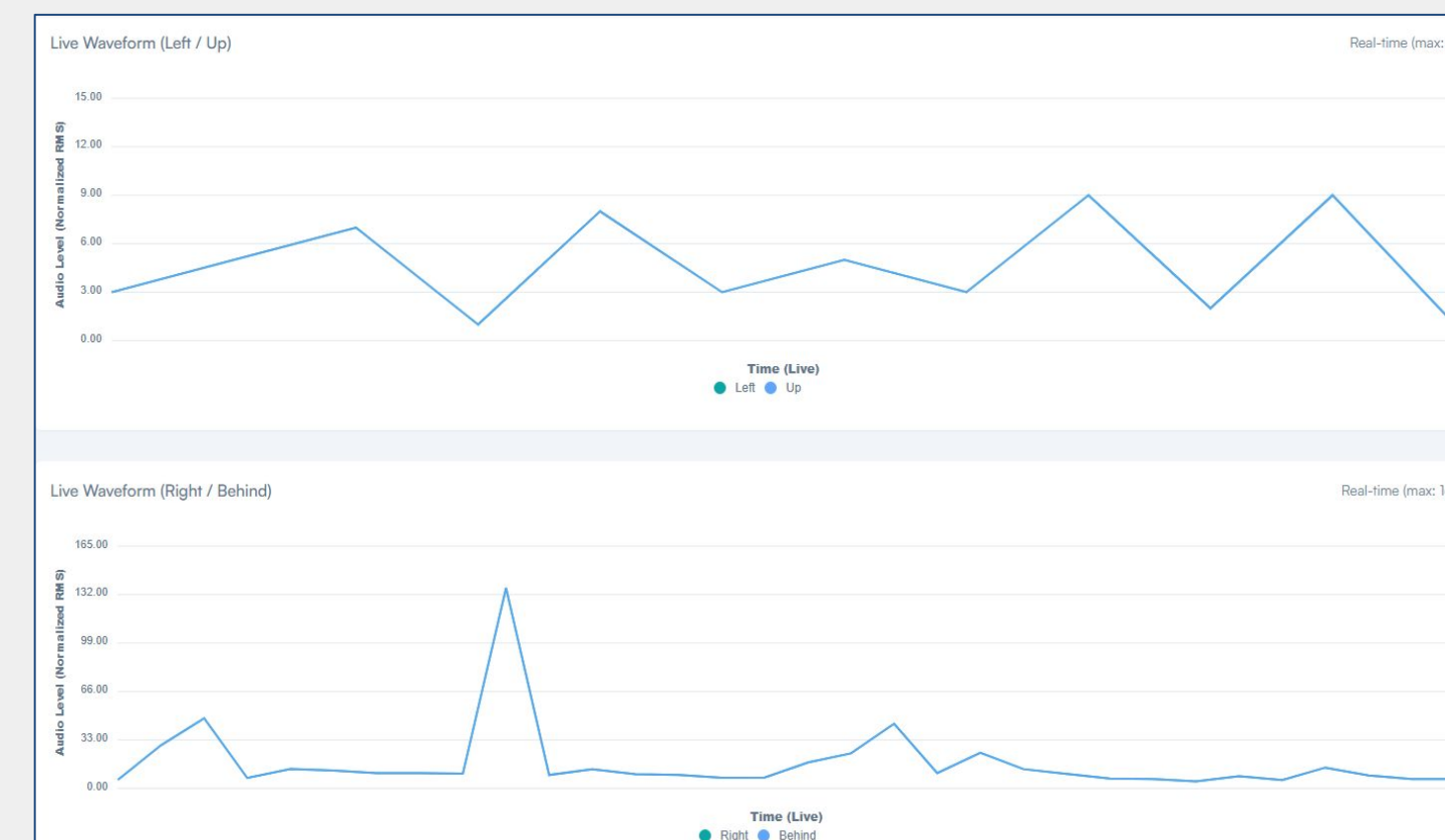


Fig 4: Prototype of Sound Reactive LEDs



Results

As a group, we have successfully mounted an audio-to-visual mechanism, made out of leds, microphones, and two microcontrollers, onto a pair of wearable glasses. Additionally, we were able to map different leds on our glasses to flash to a certain color depending on where an “abrupt” sound is coming from. Through the use of threshold frequencies, we were able to limit all of our LEDs from going off whenever they all heard a noise, in order to stop confusion of where the sound came from. Lastly, we were also able to implement code that allowed our LEDs to measure sound at $\pm 45^\circ$, while also holding a latency response of only 200ms. We conducted 50 trials of various sound inputs, in 42 of these trials, the correct LED activated, resulting in an overall detection accuracy of 84%.

Additional Information

Electret microphones made the impact of noise extremely strong to the point where our code was unable to properly detect certain sounds that surpassed thresholds made to detect abrupt sounds. This was due to the microphones high sensitivity leading us to switch to MEMS microphones, as they proved to be better at reducing noise. Additionally, we found printing mini separate mounts for each microphone and LED inefficient, leading us to print one large mount that could fit over the entire frame of our glasses, and be used to hold our LEDs and microphones in place.

References

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- [5] Market.us Media, “Hearing Loss Statistics by Age Group, Type, Problems (2026),” Market.us, Jan. 30, 2026. <https://media.market.us/hearing-loss-statistics/>